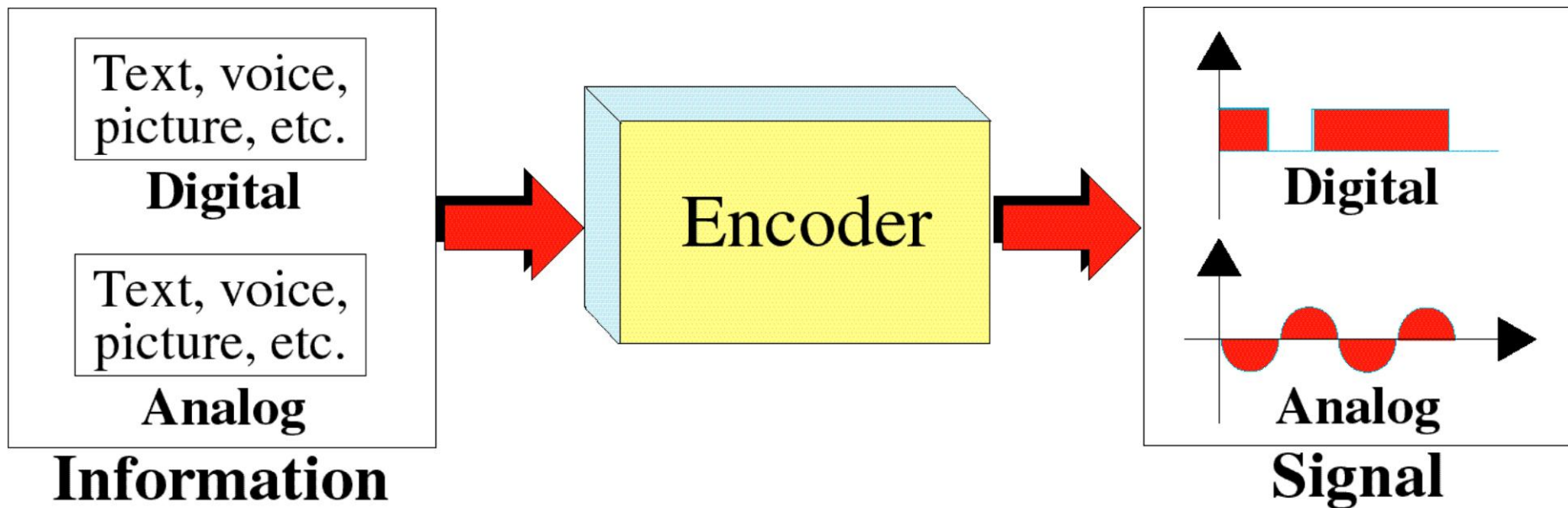


Signals

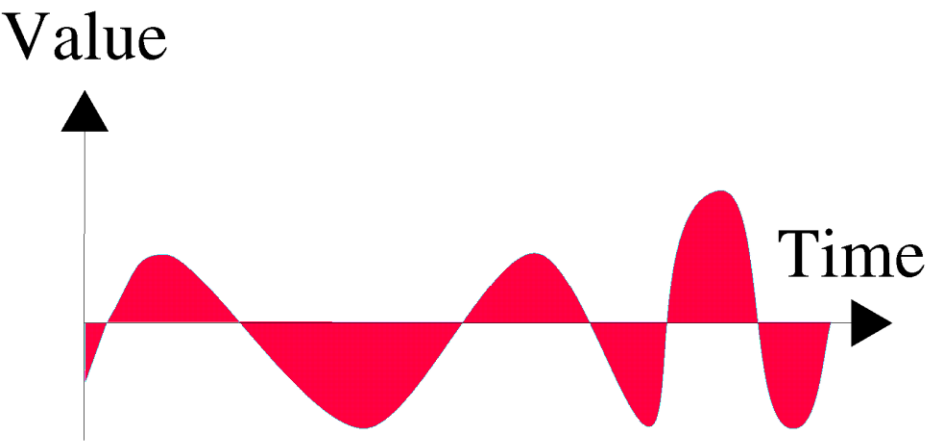
A major concern of the physical layer is moving information in the form of electromagnetic signal across a transmission medium.

- **Analog and digital**
- **Aperiodic and periodic signals**
- **Analog signals**
- **Digital signals**

Transformation of Information to Signals

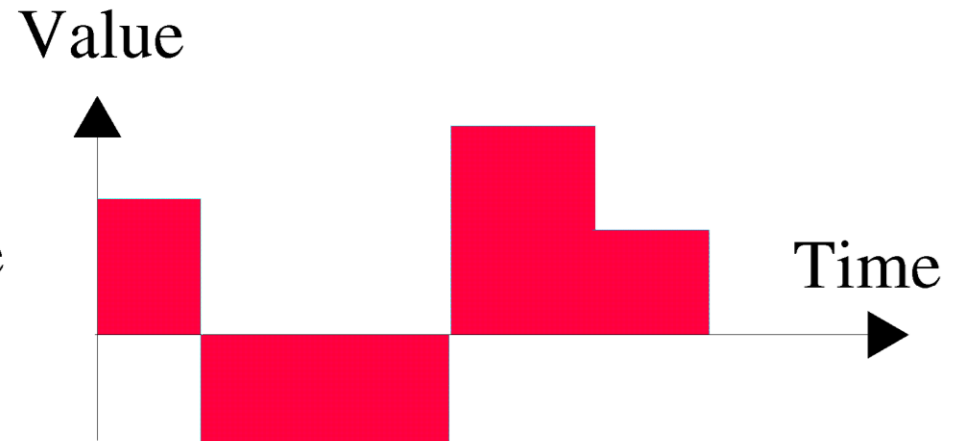


Analog and Digital Signals



a. Analog signal

Continuous wave form that changes smoothly over time



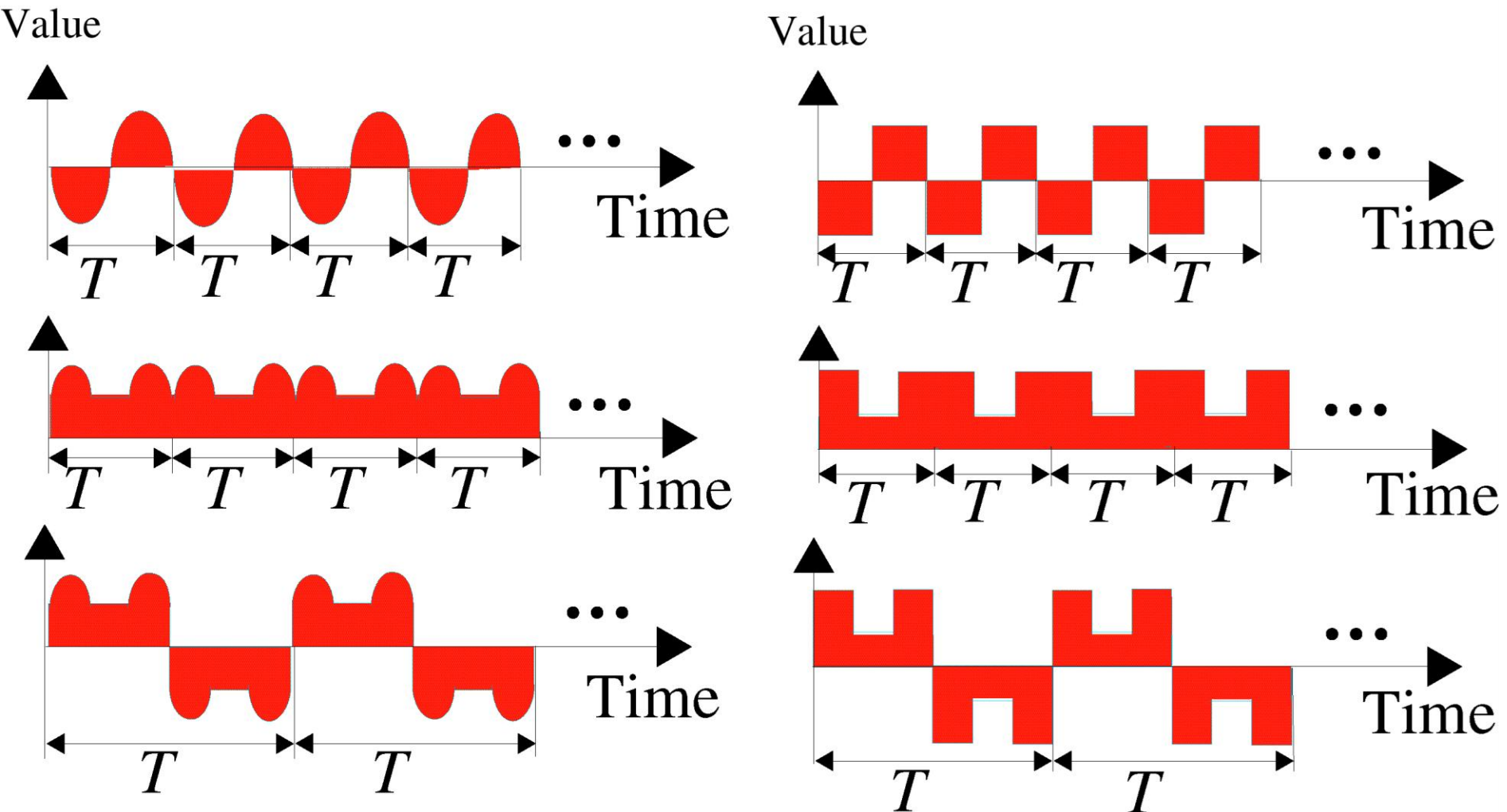
b. Digital signal

Discrete, can have only limited numbers of values.

Digital and analog signals can be of two forms:

- Periodic signal
- Aperiodic signal

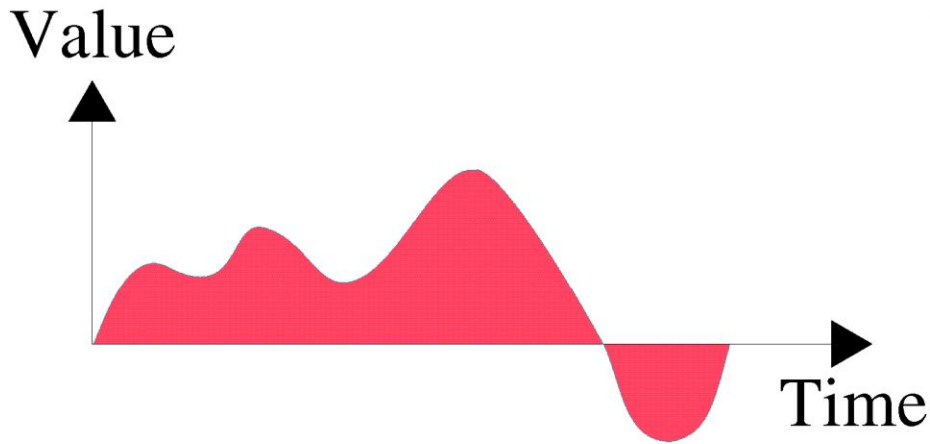
Periodic Signals: consist of continuously repeated pattern.



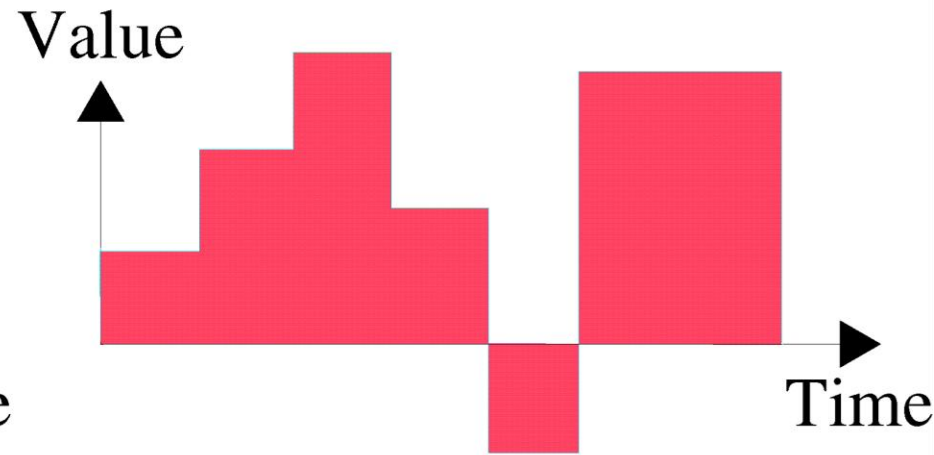
a. Analog

b. Digital

Aperiodic Signals: has no repetitive pattern.



a. Analog

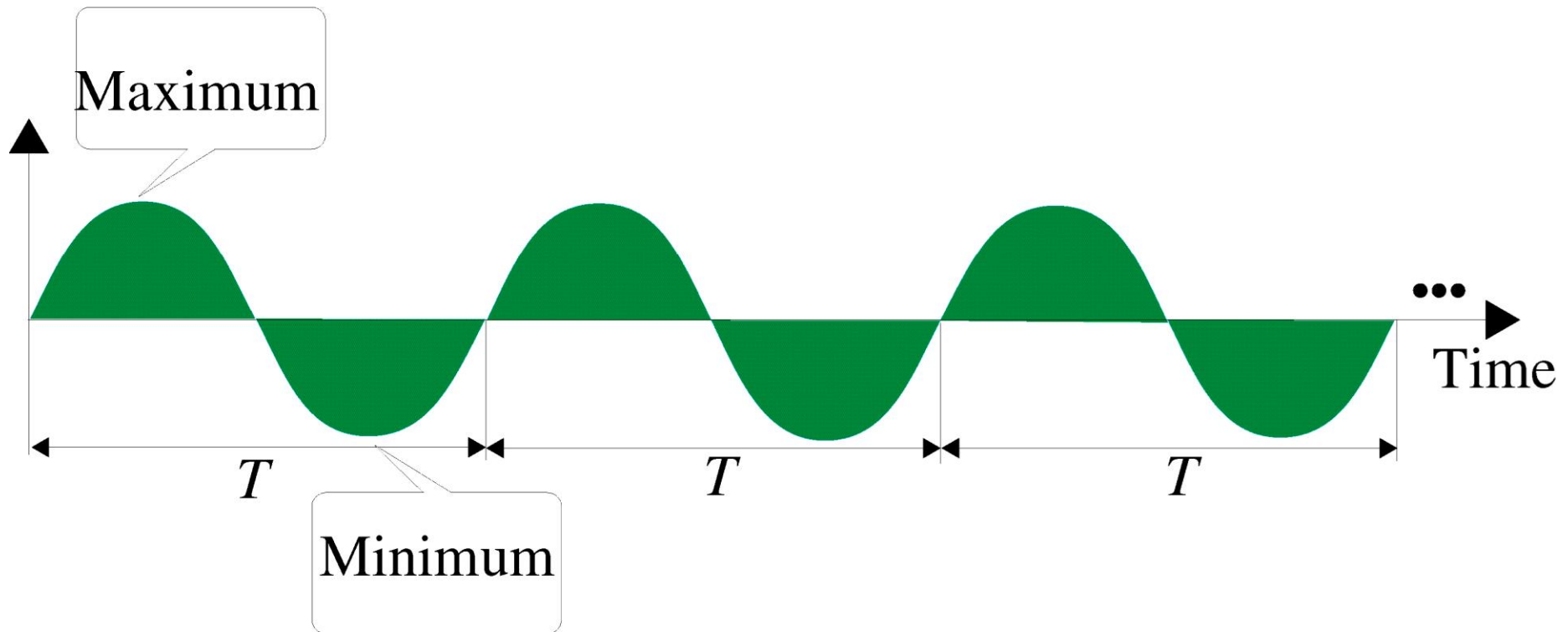


b. Digital

Any aperiodic signal can be decomposed into an infinite number of periodic signals.

Analog Signal

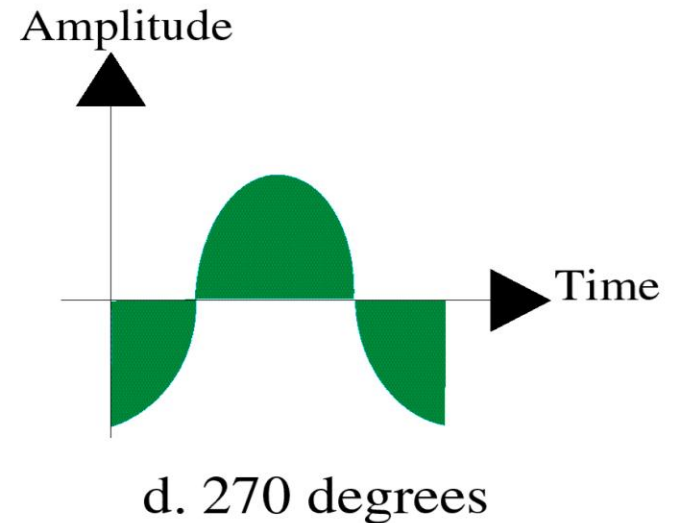
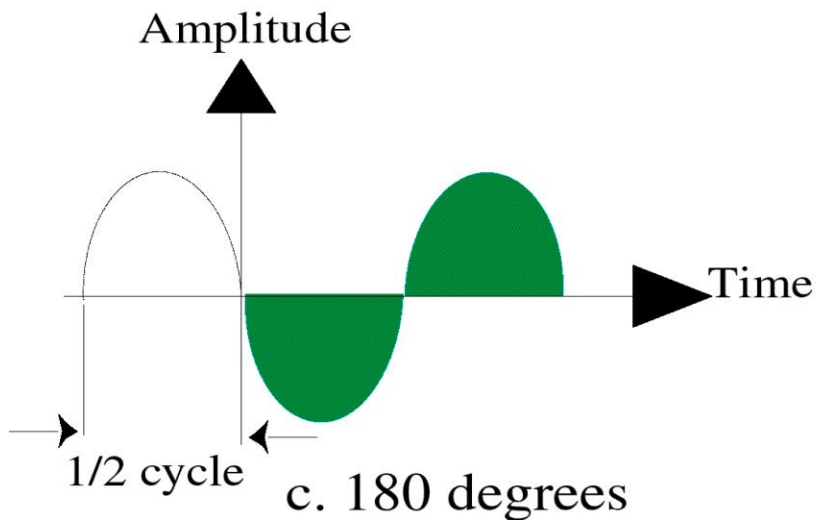
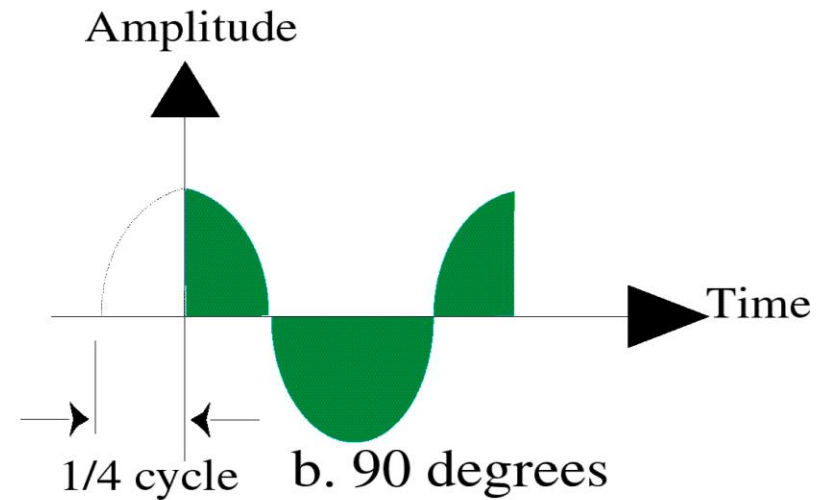
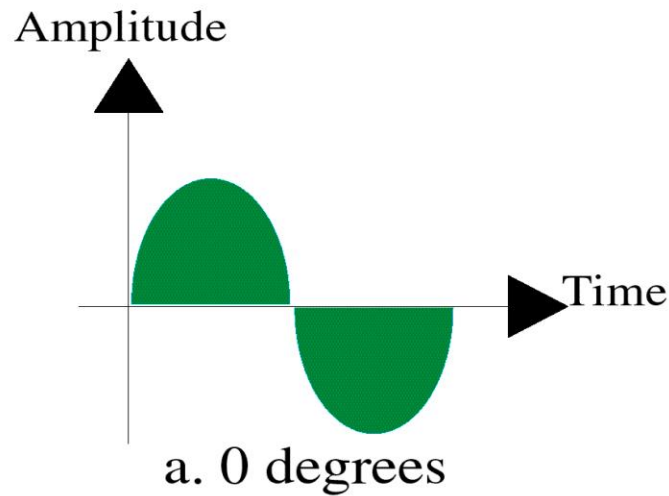
Sine Wave



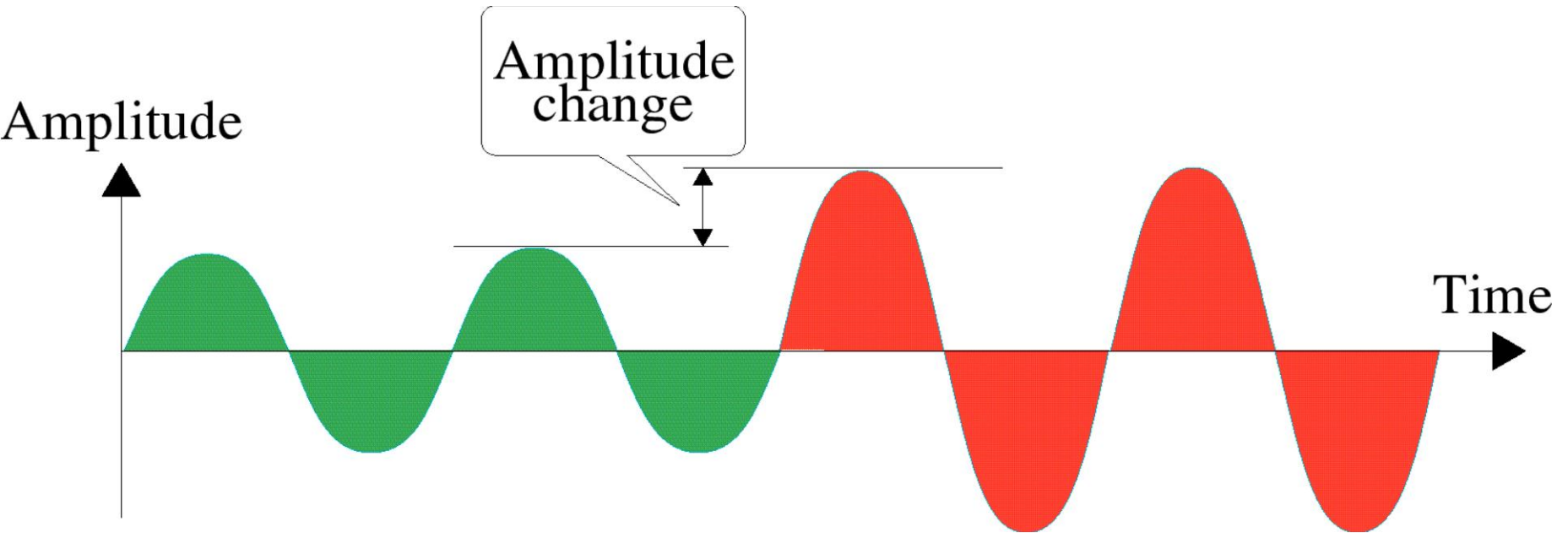
Period: amount of time to complete one cycle.

Frequency: no of cycle in 1 second.

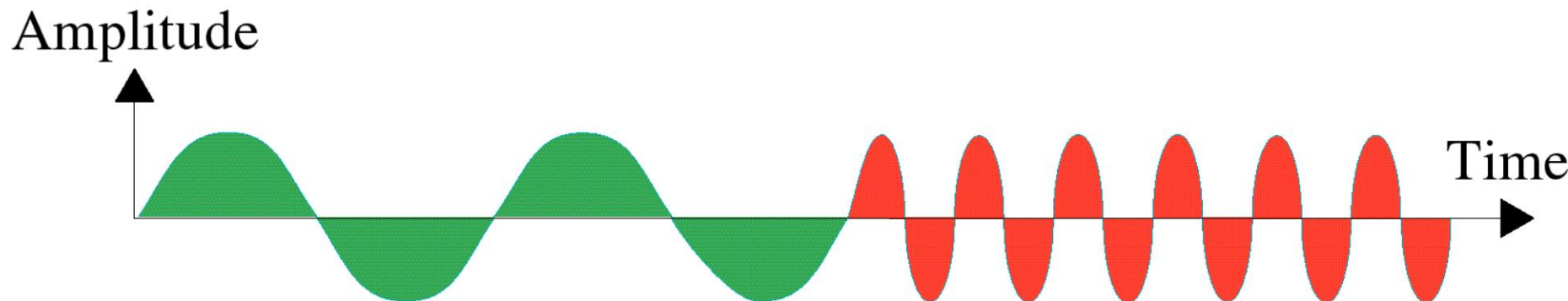
Phases: The position of the wave form relative to time zero.



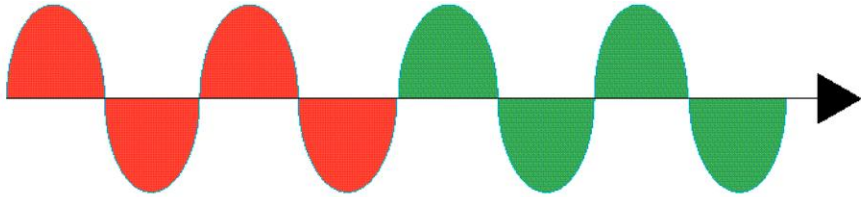
Amplitude Change



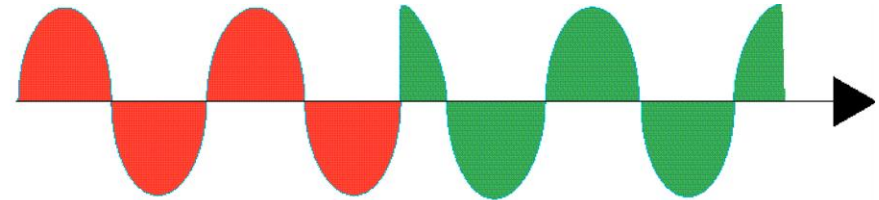
Frequency Change



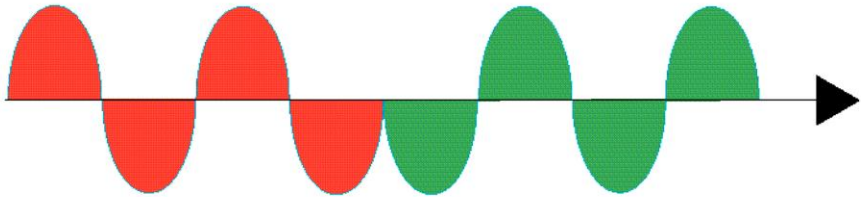
Phase Change



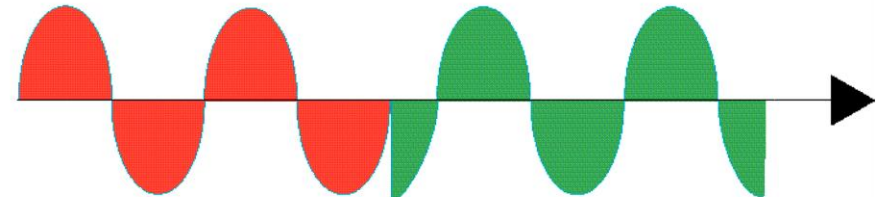
a. No phase change



b. 90 degree phase change

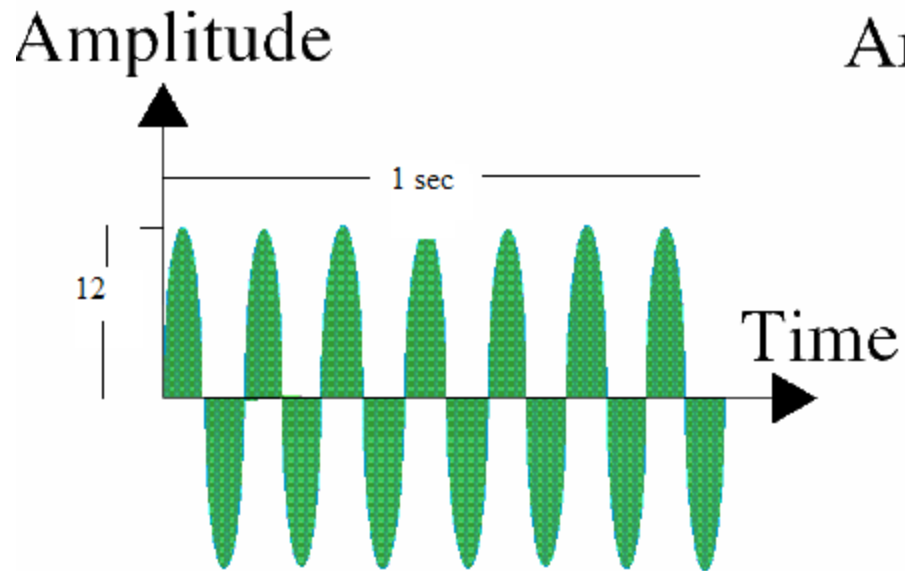


c. 180 degree phase change

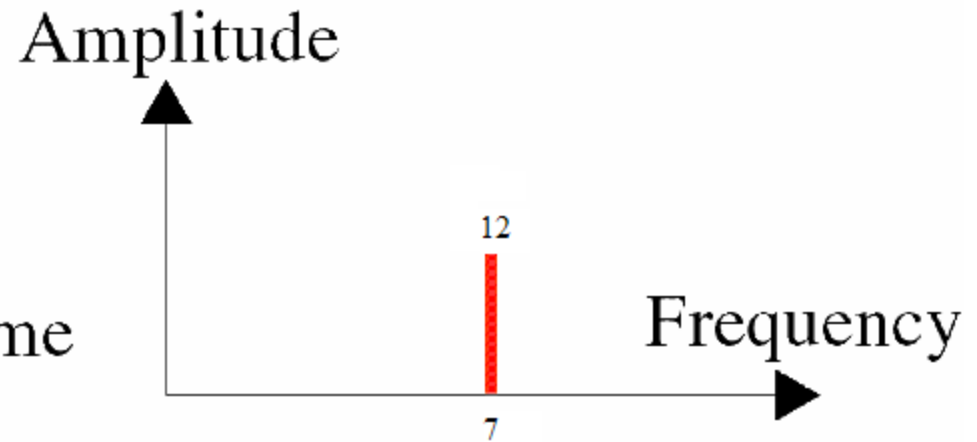


d. 270 degree phase change

Time and Frequency Domain

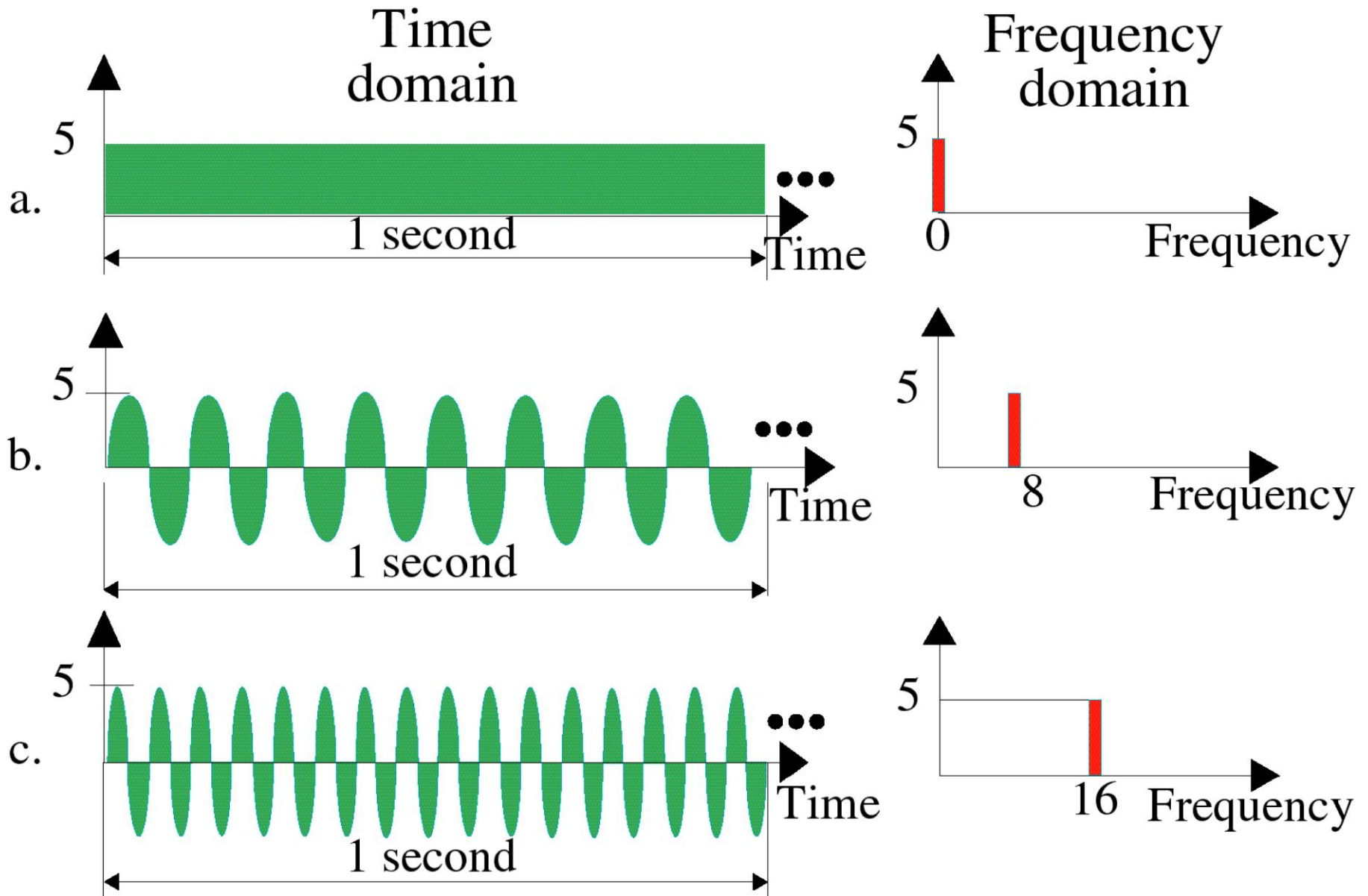


a. Time domain



b. Frequency domain

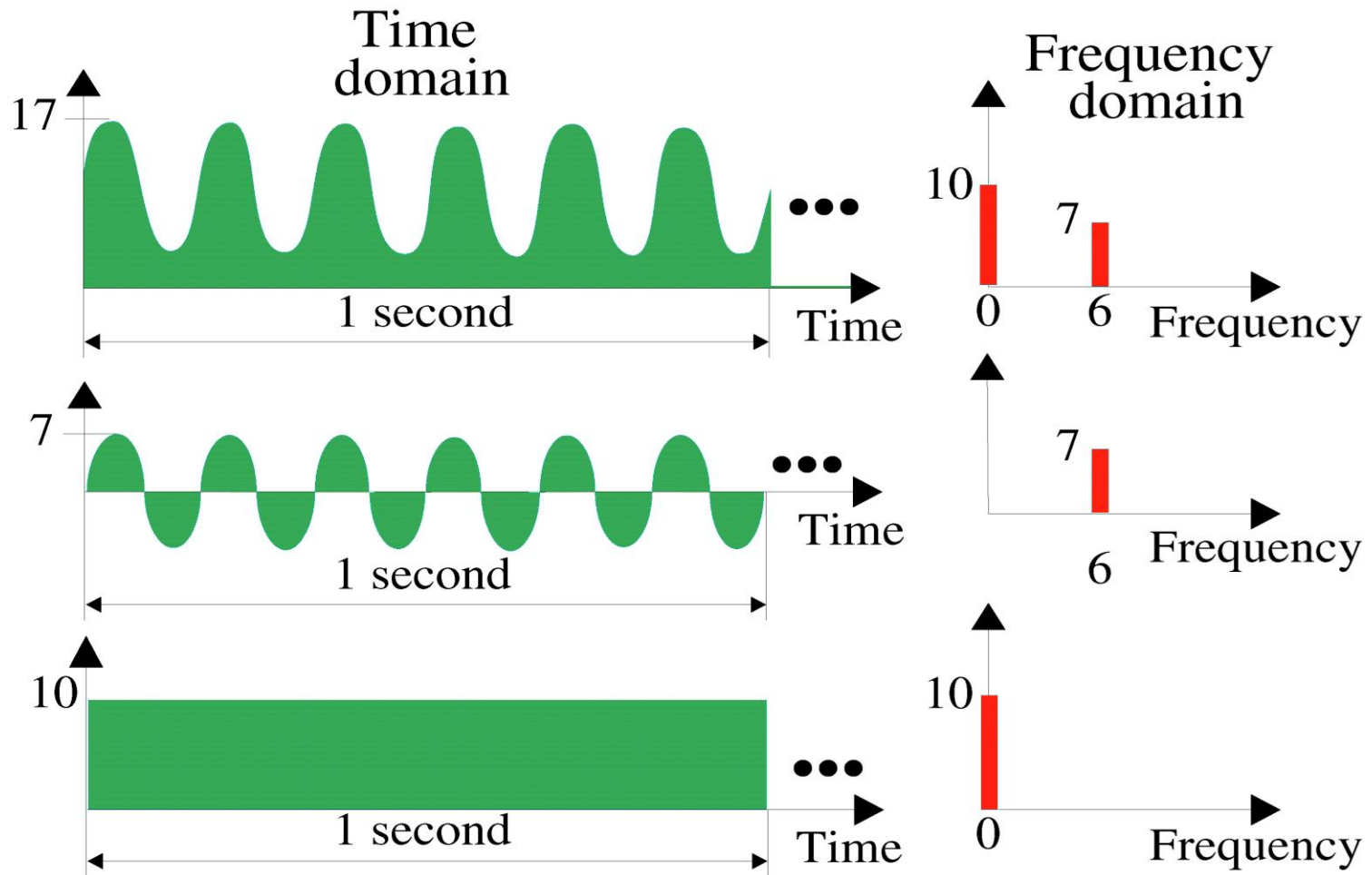
Examples



Signal with DC Component

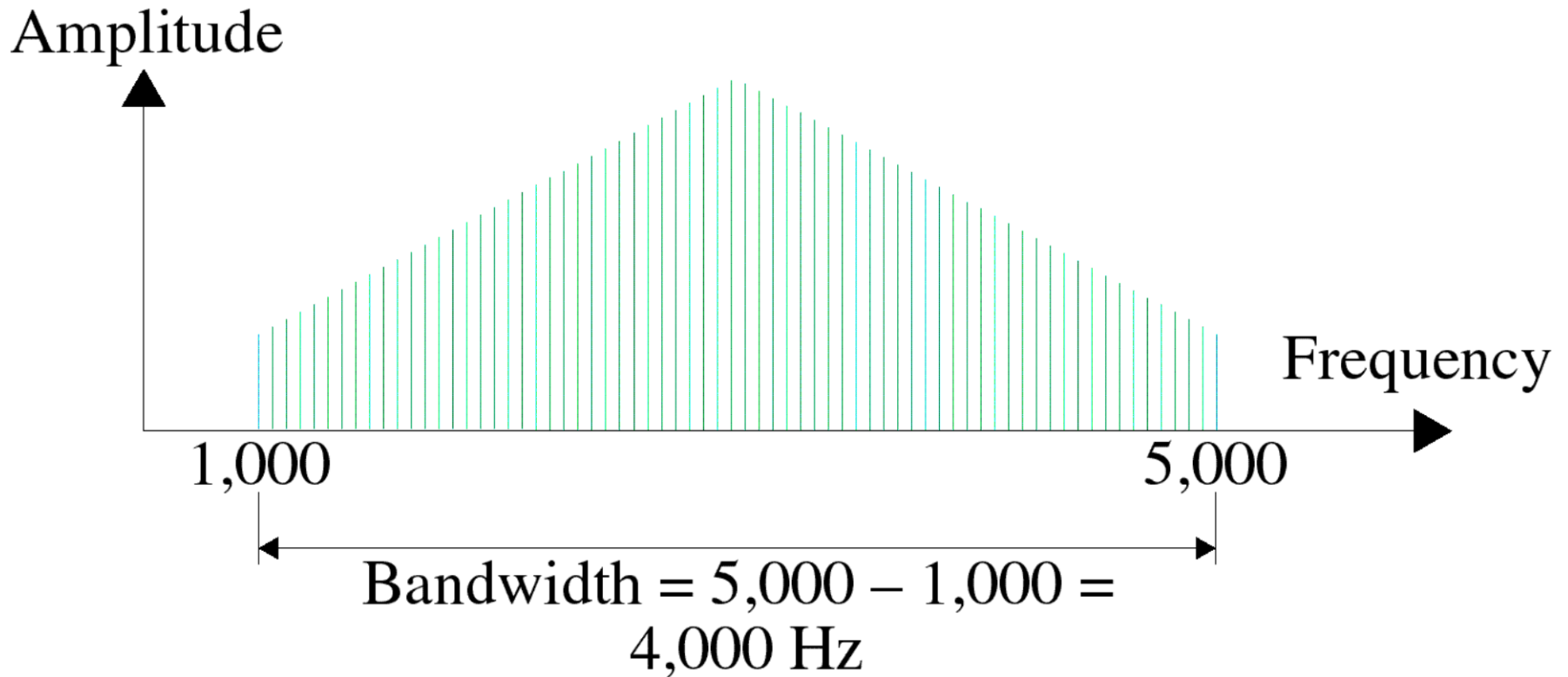
Presence of a zero frequency component. The average amplitude of this signal is non zero.

Composite signal: A periodic signal can be decomposed into a collection of sine waves.



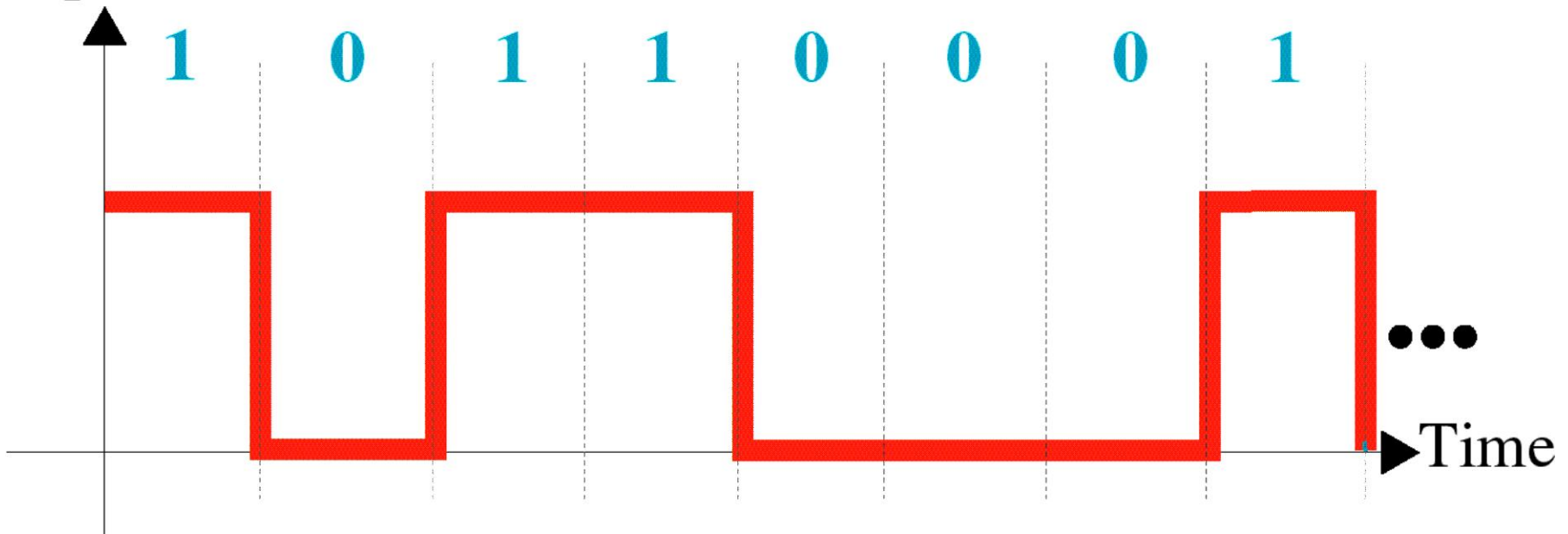
Bandwidth

Difference between lowest and highest frequency

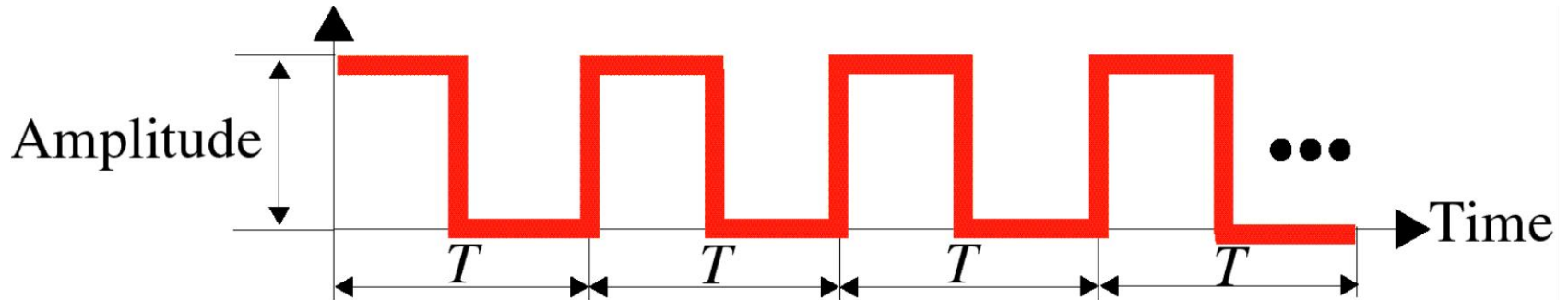


Digital Signal

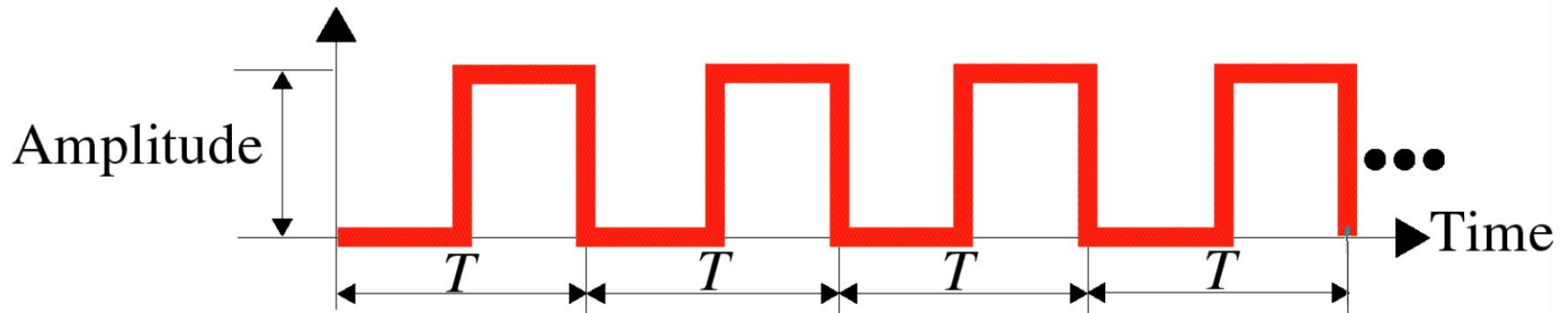
Amplitude



Amplitude, Period, and Phase for a Digital Signal



a. No phase shift



b. 180 degree phase shift

Bit Rate and Bit Interval

